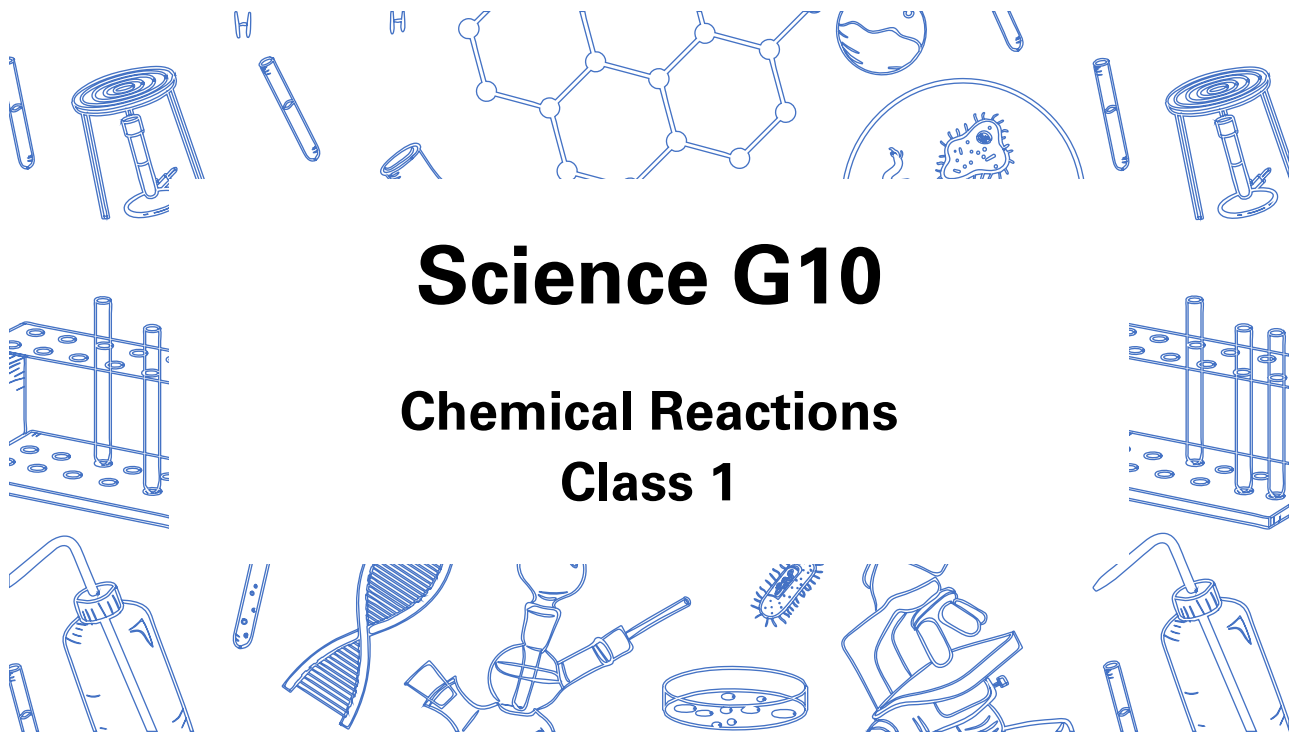




Science G10

Chemical Reactions Class 1

Agenda

1. Course Outline
2. Classkick Portfolio Account Setup
3. Diagnostic Test
4. Lesson 1
 - Quick Review of G9 Chemistry
 - Lewis Dot Diagrams
 - Ionic Compounds & Nomenclature
 - Covalent Bonding & Nomenclature



Chemistry

- **Chemistry** is the study of matter and the changes it undergoes
- **Matter** is anything that has mass and volume



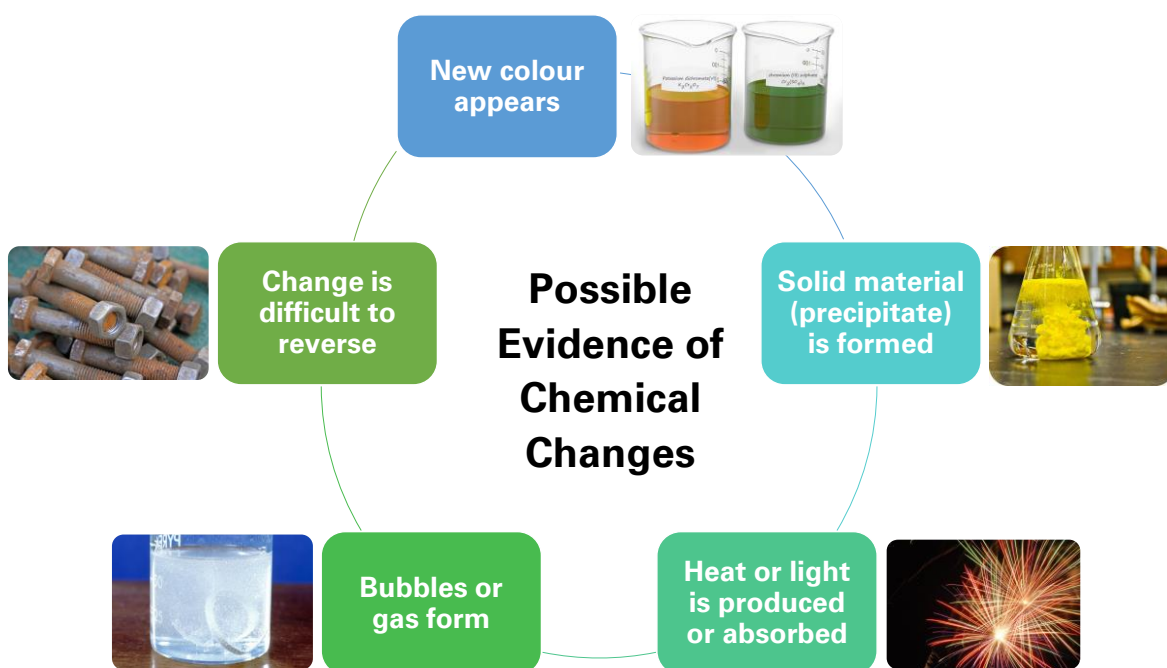
Physical & Chemical Properties

Physical Property can be measured and observed without changing the composition or identity of a substance

- Colour
- Density
- Solubility
- Melting point

Chemical Property can be observed when a substance changes into one or more new substances

- Flammability
- Bleaching
- Corrosion
- Rusting

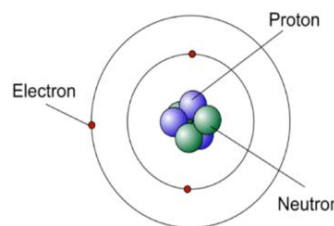


Checkpoint

Classify the following as a physical or chemical property:

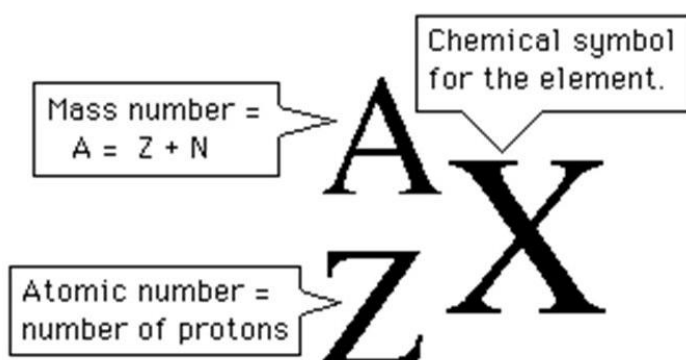
- a) Liquid nitrogen boils at -196°C
- b) Propane, leaking from a tank, ignites easily
- c) Silver jewelry tarnishes in air
- d) Spilled oil floats on the surface of water
- e) Meat darkens when it is heated on a grill
- f) Sulfur trioxide changes to sulfuric acid in the atmosphere

Atomic Structure



- An atom consists of:

	Proton	Neutron	Electron
Electrical Charge	+	0	-
Relative Mass	1	1	1/2000
Symbol	p^+	n^0	e^-
Location	Nucleus	Nucleus	Outside Nucleus



- **Mass Number (A)** is the total number of neutrons and protons present in the nucleus of an atom of an element
- **Atomic Number (Z)** is the number of protons in the nucleus of each atom of an element

Terms to Know

- **Isotopes** are atoms with the same atomic number but different mass numbers
- **Ions** are atoms with the same atomic number but a different number of electrons
 - **Cation** is a positively-charged ion
 - **Anion** is a negatively-charged ion
 - **Note: Atoms become ions when there is a loss/gain of **electrons** NOT protons**

	Li-6	Li-7
Protons	3	3
Electrons	3	3
Neutrons	3	4

	Na	Na ⁺
Protons	11	11
Electrons	11	10
Neutrons	12	12



Checkpoint



List the number of p^+ , n^0 , and e^- for:

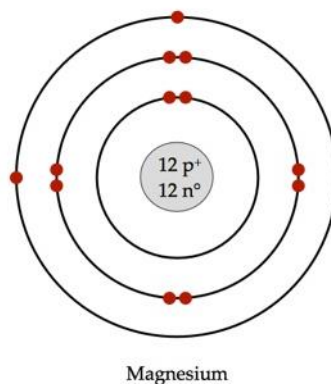
a) Beryllium

b) Phosphorus

c) Fluoride ion (F^-)

Bohr-Rutherford Diagrams

1. Determine the number of protons, neutrons, and electrons
2. Draw the nucleus and write the number of protons and neutrons inside
3. Add the electrons to the outer shells:
 - 1st shell = 2 electrons
 - 2nd shell = 8 electrons
 - 3rd shell = 8 electrons
 - 4th shell = 18 electrons



Checkpoint



Draw the Bohr-Rutherford Diagram for:

a) Boron-10

b) K⁺

Patterns in the Periodic Table

- **Group/Family** is a column in the periodic table
- **Period** is a row in the periodic table

The periodic table shows elements arranged by increasing atomic number. It is divided into periods (rows) and groups (columns). The groups are color-coded as follows:

- Group 1 (IA): Red
- Group 2 (IIA): Orange
- Groups 3-10 (Transition metals): Blue
- Groups 11-12 (Post-transition metals): Purple
- Groups 13-15 (Metalloids): Yellow
- Groups 16-17 (Nonmetals): Green
- Group 18 (Noble gases): Pink
- Lanthanides (bottom left): Light blue
- Actinides (bottom right): Dark blue
- Unknown chemical properties (bottom right): Grey

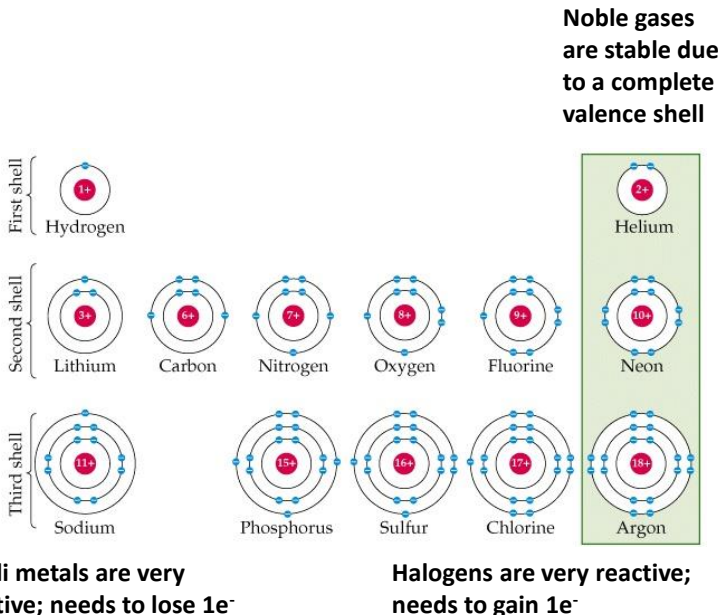
	Alkali metals
	Alkaline earth metals
	Transition metals
	Post-transition metals
	Metalloids
	Nonmetals
	Noble gases
	Lanthanides
	Actinides
	Unknown chemical properties

Trends in the Periodic Table

- Atomic number steadily increases across a period
- Number of energy levels (shells) steadily increases down a group
- Metals tend to lose electrons
- Nonmetals tend to attract electrons
- Number of valence electrons (outermost electrons) are the same within a group and increase steadily across a period

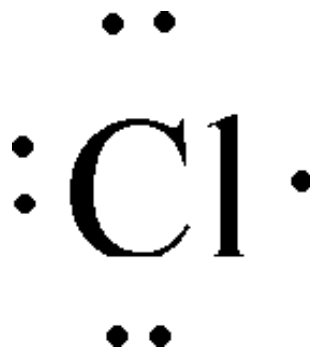
Reactivity

- Valence electrons determine the reactivity of an element and how compounds are formed
- Elements tend to lose or gain valence electrons to form bonds and achieve stability



Lewis Dot Diagrams

- Write the element symbol
- Find the number of valence electrons by looking at the group number. For Groups 13-18, subtract 10 to obtain the valence electrons
- Place dots representing the valence electrons on the four sides of the element symbol singly before pairing up



Checkpoint

Draw the Lewis Dot Diagram of:

a) Silicon

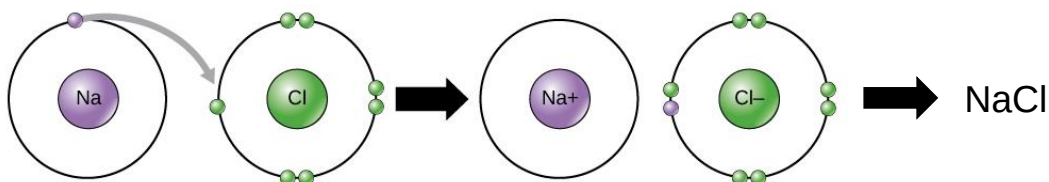
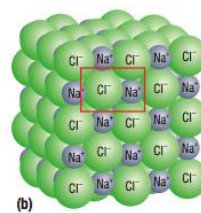
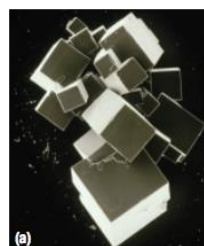
b) Ca

c) Mg^{2+}

d) O^{2-}

Ionic Compounds

- **Ionic Compounds** are formed by the electrostatic interaction of a cation and an anion; metal and a nonmetal



Periodic Table of the Elements

GROUP 1

Atomic Number, Symbol, Name, Atomic Weight, Electronegativity, Possible Ionic Charges (First charge is more stable)

Period

18

He Helium

19 K Potassium

20 Ca Calcium

21 Sc Scandium

22 Ti Titanium

23 V Vanadium

24 Cr Chromium

25 Mn Manganese

26 Fe Iron

27 Co Cobalt

28 Ni Nickel

29 Cu Copper

30 Zn Zinc

31 Ga Gallium

32 Ge Germanium

33 As Arsenic

34 Se Selenium

35 Br Bromine

36 Kr Krypton

37 Rb Rubidium

38 Sr Strontium

39 Y Yttrium

40 Zr Zirconium

41 Nb Niobium

42 Mo Molybdenum

43 Tc Technetium

44 Ru Ruthenium

45 Rh Rhodium

46 Pd Palladium

47 Ag Silver

48 Cd Cadmium

49 In Indium

50 Sn Tin

51 Sb Antimony

52 Te Tellurium

53 I Iodine

54 Xe Xenon

55 Cs Cesium

56 Ba Barium

57 La Lanthanum

58 Ce Cerium

59 Pr Praseodymium

60 Nd Neodymium

61 Pm Promethium

62 Sm Samarium

63 Eu Europium

64 Gd Gadolinium

65 Tb Terbium

66 Dy Dysprosium

67 Ho Holmium

68 Er Erbium

69 Tm Thulium

70 Yb Ytterbium

71 Lu Lutetium

72 Hf Hafnium

73 Ta Tantalum

74 W Tungsten

75 Re Rhenium

76 Os Osmium

77 Ir Iridium

78 Pt Platinum

79 Au Gold

80 Hg Mercury

81 Tl Thallium

82 Pb Lead

83 Bi Bismuth

84 Po Polonium

85 At Astatine

86 Rn Radon

87 Fr Francium

88 Ra Radium

89 Ac Actinium

90 Th Thorium

91 Pa Protactinium

92 U Uranium

93 Np Neptunium

94 Pu Plutonium

95 Am Americium

96 Cm Curium

97 Bk Berkelium

98 Cf Californium

99 Es Einsteinium

100 Fm Fermium

101 Md Mendelevium

102 No Nobelium

103 Lr Lawrencium

104 Rf Rutherfordium

105 Db Dubnium

106 Sg Seaborgium

107 Bh Bohrium

108 Hs Hassium

109 Mt Meitnerium

110 Ds Darmstadtium

111 Rg Roentgenium

112 Cn Copernicium

113 Nh Nihonium

114 Fl Flerovium

115 Mc Moscovium

116 Lv Livermorium

117 Ts Tennessine

118 Og Oganesson

119

120

121

122

123

124

125

126

127

128

129

130

131

132

133

134

135

136

137

138

139

140

141

142

143

144

145

146

147

148

149

150

151

152

153

154

155

156

157

158

159

160

161

162

163

164

165

166

167

168

169

170

171

172

173

174

175

176

177

178

179

180

181

182

183

184

185

186

187

188

189

190

191

192

193

194

195

196

197

198

199

200

201

202

203

204

205

206

207

208

209

210

211

212

213

214

215

216

217

218

219

220

221

222

223

224

225

226

227

228

229

230

231

232

233

234

235

236

237

238

239

240

241

242

243

244

245

246

247

248

249

250

251

252

253

254

255

256

257

258

259

260

261

262

263

264

265

266

267

268

269

270

271

272

273

274

275

276

277

278

279

280

281

282

283

284

285

286

287

288

289

290

291

292

293

294

295

296

297

298

299

300

301

302

303

304

305

306

307

308

309

310

311

312

313

314

315

316

317

318

319

320

321

322

323

324

325

326

327

328

329

330

331

332

333

334

335

336

337

338

339

340

341

342

343

344

345

346

347

348

349

350

351

352

353

354

355

356

357

358

359

360

361

362

363

364

365

366

367

368

369

370

371

372

373

374

375

376

377

378

379

380

381

382

383

384

385

386

387

388

389

390

391

392

393

394

395

396

397

398

399

400

401

402

403

404

405

406

407

408

409

410

411

412

413

414

415

416

417

418

419

420

421

422

423

424

425

426

427

428

429

430

431

432

433

434

435

436

437

438

439

440

441

442

443

444

445

446

447

448

449

450

451

452

453

454

455

456

457

458

459

460

461

462

463

464

465

466

467

468

469

470

471

472

473

474

475

476

477

478

479

480

481

482

483

484

485

486

487

488

489

490

491

492

493

494

495

496

497

498

499

500

501

502

503

504

505

506

507

508

509

510

511

512

513

514

515

516

517

518

519

520

521

522

523

524

525

526

527

528

529

530

531

532

533

534

535

536

537

538

539

540

541

542

543

544

545

546

547

548

549

550

551

552

553

554

555

556

557

558

559

560

561

562

563

564

565

566

567

568

569

570

571

572

573

574

575

576

577

578

579

580

581

582

583

584

585

586

587

588

589

590

591

592

593

594

595

596

597

598

599

600

601

602

603

604

605

606

607

608

609

610

611

612

613

614

615

616

617

618

619

620

621

622

623

624

625

626

627

628

629

630

631

632

633

634

635

636

637

638

639

640

641

642

643

644

645

646

647

648

649

650

651

652

653

654

655

656

657

658

659

660

661

662

663

664

665

666

667

668

669

670

671

672

673

674

675

676

677

678

679

680

681

682

683

684

685

686

687

688

689

690

691

692

693

694

695

696

697

698

699

700

701

702

703

704

705

706

707

708

709

710

711

712

713

714

715

716

717

718

719

720

721

722

723

724

725

726

727

728

729

730

731

732

733

734

735

736

737

738

739

740

741

742

743

744

745

746

747

748

749

750

751

752

753

754

755

756

757

758

759

760

761

762

763

764

765

766

767

768

769

770

771

772

773

774

775

776

777

778

779

780

781

782

783

784

785

786

787

788

789

790

791

792

793

794

795

796

797

798

799

800

801

802

803

804

805

806

807

808

809

810

811

812

813

814

815

816

817

818

819

820

821

822

823

824

825

826

827

828

829

830

831

832

833

834

835

836

837

838

839

840

841

842

843

844

845

846

847

848

849

850

851

852

853

854

855

856

857

858

859

860

861

862

863

864

865

866

867

868

869

870

871

872

873

874

875

876

877

878

879

880

881

882

883

884

885

886

887

888

889

890

891

892

893

894

895

896

897

898

899

900

901

902

903

904

905

906

907

908

909

910

911

912

913

914

915

916

917

918

919

920

921

922

923

924

925

926

927

928

929

930

931

932

933

934

935

936

937

938

939

940

941

942

943

944

945

946

947

948

949

950

951

952

953

954

955

956

957

958

959

960

961

962

963

964

965

966

967

968

969

970

971

972

973

974

975

976

977

978

979

980

981

982

983

984

985

986

987

988

989

990

991

992

993

994

995

996

997

998

999

1000

Metals tend to lose valence electrons

Non-metals tends to gain valence electrons

Physical Properties of Ionic Compounds

- High melting points and boiling points
- Low volatility/Non-volatile
- Generally soluble in polar solvents
- Do not conduct electricity in the solid state
- Conduct electricity when molten or when dissolved in water
- Generally brittle

Ionic compound	Charge on metal ion	Melting point
Na ₂ O	1+	1132 °C
MgO	2+	2800 °C

Writing Chemical Formulas

Crisscross Method

Ex: Calcium chloride

1. Write the element symbols of the metal and then the nonmetal
2. Find the ionic charge of each element on the periodic table and write the ionic charge on the top-right corner
3. Criss-cross the ionic charge
4. Simplify the subscripts to the lowest terms and remove the charges



Checkpoint



Draw the Lewis Dot Diagram for the formation of the following:

- a) Magnesium bromide
- b) Boron nitride
- c) Aluminum fluoride



Checkpoint



Write the chemical formula for the following:

- a) Magnesium bromide
- b) Boron nitride
- c) Aluminum fluoride

Writing Chemical Names

1. Un-crisscross the charges and consider if the compound was simplified.
2. Write the name of the metal and check its ionic charge on the periodic table
3. Write the name of the nonmetal and check its ionic charge on the periodic table. For anions, change the ending to *-ide*

Ex: ~~BeO~~

Be¹⁺ O¹⁻

Be²⁺ O²⁻

Beryllium oxygen

Beryllium oxide

Multivalent Elements

- Some transition metals are multivalent (more than one ionic charge)
- Use Roman numerals to distinguish between the ionic charges

1A	2																	8A
	2A																	
Li ⁺																		
Na ⁺	Mg ²⁺	3	4	5	6	7	8	9	10	11	12							
		3B	4B	5B	6B	7B	8B											
K ⁺	Ca ²⁺				Cr ²⁺	Mn ²⁺	Fe ²⁺	Co ²⁺	Ni ²⁺	Cu ⁺	Zn ²⁺							
					Cr ³⁺	Mn ³⁺	Fe ³⁺	Co ³⁺	Ni ³⁺	Cu ²⁺								
Rb ⁺	Sr ²⁺									Ag ⁺	Cd ²⁺							
Cs ⁺	Ba ²⁺									Au ⁺	Hg ²⁺							
										Au ³⁺	Hg ²⁺							

Ion	Chemical Name
Fe ²⁺	Iron(II)
Fe ³⁺	Iron(III)
Cu ⁺	Copper(I)
Cu ²⁺	Copper(II)



Checkpoint



Write the chemical formula for the following:

- Copper(II) chloride
- Iron(III) bromide
- Nickel(II) oxide

Checkpoint

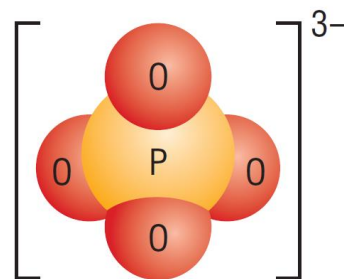
Write the chemical name for the following:



Polyatomic Ions

- **Polyatomic ion** is made of more than one atom and acts as a single unit

Ex: PO_4^{3-} phosphate ion
 OH^- hydroxide ion
 CO_3^{2-} carbonate ion



TIP: It may be helpful to keep the elements within the parentheses and the charge outside the parentheses.

Ex: $(\text{PO}_4)^{3-}$ or $(\text{OH})^-$

Writing Chemical Formulas with Polyatomic Ions

1. Write the metal with its ionic charge
2. Write the polyatomic ion with its charge outside the parentheses
3. Crisscross the ionic charges
4. Simplify the subscripts
 - Do not simplify the elements within the parentheses

Ex: Iron(III) nitrate



Naming Chemical Names with Polyatomic Ions

1. Place parentheses around the polyatomic ion
2. Un-crisscross the charges
3. Write the name of the metal and check its ionic charge on the periodic table
4. Write the name of the polyatomic ion and check its ionic charge

Ex: Na_2CO_3



Sodium carbonate

Checkpoint

Write the chemical formula for the following:

- a) Calcium sulfate
- b) Calcium chlorate
- c) Iron(III) phosphate

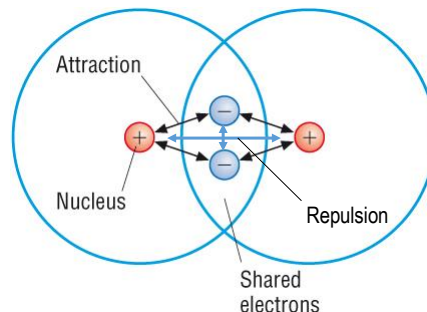
Checkpoint

Write the chemical name for the following:

- a) KNO_3
- b) $(\text{NH}_4)_3\text{PO}_4$
- c) PbCO_3

Covalent Bonding

- **Covalent bonding** is the stable balance of attractive forces and repulsive forces between atoms
 - Formed between nonmetals



a) Nitrous oxide (N_2O) is commonly known as laughing gas



b) Nitrogen dioxide (NO_2) is the reddish-brown gas in smog

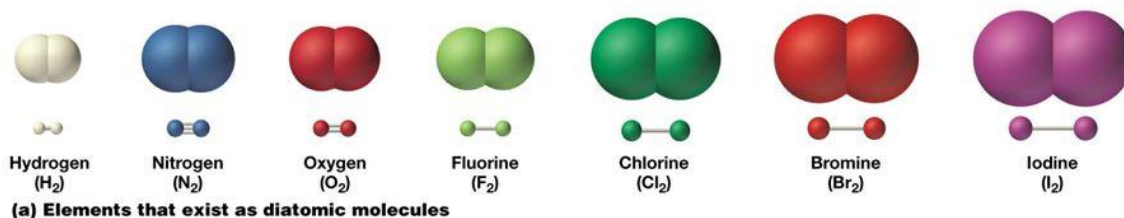
Physical Properties of Covalent Compounds

- Lower melting points and boiling points than ionic compounds
- Some are volatile (i.e. F_2 , Cl_2 , Br_2)
- Low solubility in water
- Do not usually conduct electricity as a solid, liquid, or in the aqueous state



Diatomic Molecules

- **Diatomic molecules** are molecules that consist of two atoms joined with covalent bonds
- Remember: H_2 O_2 F_2 Br_2 I_2 N_2 Cl_2 *Also P_4 and S_8



Naming Covalent Molecules

1. Write the name of the elements
2. Change the ending of the second element to *-ide*
3. Add prefixes to represent the number of atoms
 - If mono- is the first prefix, it is understood and not written
 - If the element starts with a vowel, drop the (a) on the prefix

Ex: CO_2

carbon oxygen

carbon oxide

carbon dioxide

Number	Prefix
1	Mon(o)-
2	Di-
3	Tri-
4	Tetr(a)-
5	Pent(a)-
6	Hex(a)-
7	Hept(a)-
8	Oct(a)-
9	Non(a)-
10	Dec(a)-

Checkpoint

Write the chemical names for the following:

a) CO

b) PF₅

c) N₂O

Writing Molecular Formulas

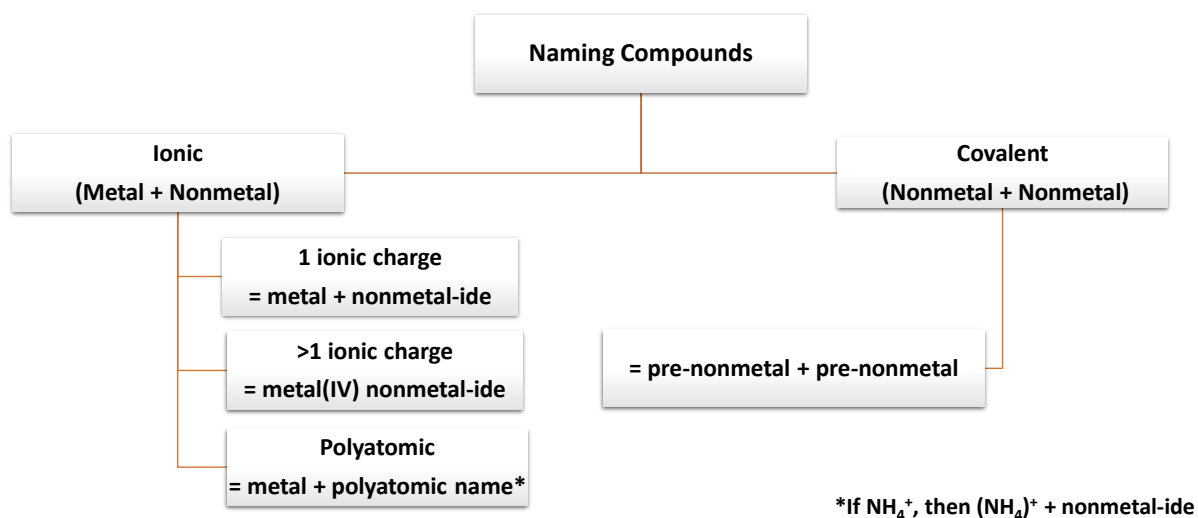
1. Write the element symbol Ex: sulfur dioxide
2. Refer to the prefix and add subscripts to the element symbol
S O
S₁ O₂
3. Do not simplify for covalent molecules
SO₂

Checkpoint

Write the chemical formulas for the following:

- a) Sulfur tetroxide
- b) Dihydrogen dioxide
- c) Carbon disulfide

Summary: Nomenclature



What I Learned Today:

- ☐ Lewis Dot Diagrams
- ☐ Physical Properties of Ionic Compounds
- ☐ Naming and Writing Ionic Compounds with:
 - Single charge
 - Multivalent charges
 - Polyatomic ions
- ☐ Naming and Writing Covalent Molecules

Due next class: Class 1 Homework

